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1. Lecture 2 in MATLAB

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Recitation due Jan 24, 2024 20:30 IST Completed

Picking out elements of a matrix, review

1/1 point (graded)

Which of the following MATLAB commands picks out the 3rd column of a matrix **A**?

☐ A(3,3)

☐ A(3,:)

☒ A(:,3)

☐ A(:,,:)



Submit

You have used 1 of 3 attempts

Find the nullspace of the matrix (External resource)

(0.0 / 1.0 points)

Nullspace problem.

The matrix **B** has the reduced row echelon form given below.

$$\text{rref}(\mathbf{B}) = \begin{pmatrix} 0 & 1 & 3 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Without using the MATLAB command null, find a basis for the nullspace of **B**. Create a vectors in that basis.

Script ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 B = [0 1 3 0 0 5; 0 0 0 1 0 7; 0 0 0 0 1 9; 0 0 0 0 0 0];
2 % Create a matrix N whose columns are a basis for the nullspace of
3 N = [1 0 0 0 0 0; 0 0 1 0 0 0; 0 0 0 0 0 1]';
```

Run Script

?

Assessment: All Tests Passed

Submit

?

Check that null is not in solution

Calculator

- ✓ Check size of N
- ✓ Check rref(B)*N is zero matrix
- ✓ Check that columns of N form a basis for

1. Lecture 2 in MATLAB

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